HPML Final Project · Spring 2026

# KV Cache Management for LLM Inference: A Comparative Study under Realistic Serving Workloads

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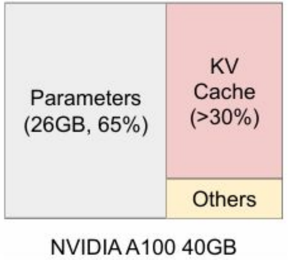
GitHub: https://github.com/Willkczy/HPML-KV-Cache-Mangement
Tracking:

*Dr. Kaoutar El Maghraoui · Columbia University · Week of May 4, 2026*

# Motivation & Problem

*Time budget: 1 minute.*

### Why this matters




KV cache scales linearly with context length and batch size, making it the dominant GPU memory bottleneck in LLM serving

At longer token contexts, the naive full-cache strategies exhaust GPU DRAM, forcing smaller batches and higher cost per token

Naive allocation causes severe memory fragmentation, limiting concurrent request throughput

### Problem statement

Serving Qwen2.5-7B under long-context workloads on a single GPU: KV cache memory pressure degrades throughput and increases latency as context grows.

**Goal:** Quantify the memory–latency–quality trade-off across four KV cache strategies on identical hardware, targeting 40–60% memory reduction with controlled quality degradation.

# Technical Approach — Model & Data

*Time budget: ~1 min of the 3-min approach section.*

### Model

Qwen2.5-7B: decoder-only, 28 layers, GQA (4 KV heads / 28 Q heads), hidden size 3584, head dim 128, bfloat16, 128k context window

Full Cache & eviction methods via Hugging Face Transformers; Page Attention via vLLM

### Dataset

| Bucket | Dataset | Tokens | Generation | Quality |
|---|---|---|---|---|
| Short | MMLU | 128-512 | 10 tok | Accuracy |
| Long QA | LongBench v2 | 8-16k | 10 tok | Accuracy |
| Long explain | LongBench v2 | 8-16k | 512 tok | Accuracy |
| Summarization | GovReport | 4-16k | 512 tok | ROUGE-L |

### Hardware

**Insomnia cluster:**

NVIDIA L40S

RTXA6000 (48 GB VRAM)

# Technical Approach — Four KV cache management strategies

1. **Full Cache:** Baseline, keeps every key and value vector for every token in the context
2. **H2O:** Keep the most important tokens instead of the full cache.
3. **StreamingLLM:** Simple sink-plus-window strategy
4. **PageAttention:** vLLM’s memory management strategy

# Technical Approach — Experiments Setups

## Experiment 1 - controlled single-request benchmark

isolate the behavior of each strategy without interference from batching, queueing, or serving scheduler effects

- MMLU short: 5-shot MMLU, 128-512 tokens, generate 10 tokens
- LongBench short answer: 8k-16k tokens, generate answer as A/B/C/D
- LongBench explain: 8k-16k tokens, generate 512 tokens, let the model explain first then give answer as A/B/C/D -> measured with accuracy
- GovReport summarization: 4k-16k tokens, generate 512 tokens on summarization -> measured with ROUGE-L

**Each method receives the same formatted prompt and returns the same output schema**

## Realistic Serving - realistic Concurrent Serving Benchmark

measure what happens when requests arrive over time and multiple requests are active simultaneously

- short: 50%, MMLU, 0-512 tokens, 8 output tokens
- medium: 30%, GovReport, 1k-4k tokens, 256 output tokens
- long: 15%, GovReport, 4k-16k tokens, 512 output tokens
- very_long: 5%, LongBench, 16k-32k tokens, 10 output tokens

**Total: 500 requests, Poisson arrivals, seed=42. Arrival rate swept: 0.5 / 1.0 / 2.0 / 4.0 req/s. Using vLLM’s AsyncLLMEngine**

# Measured Performance Metrics

**Experiment 1**

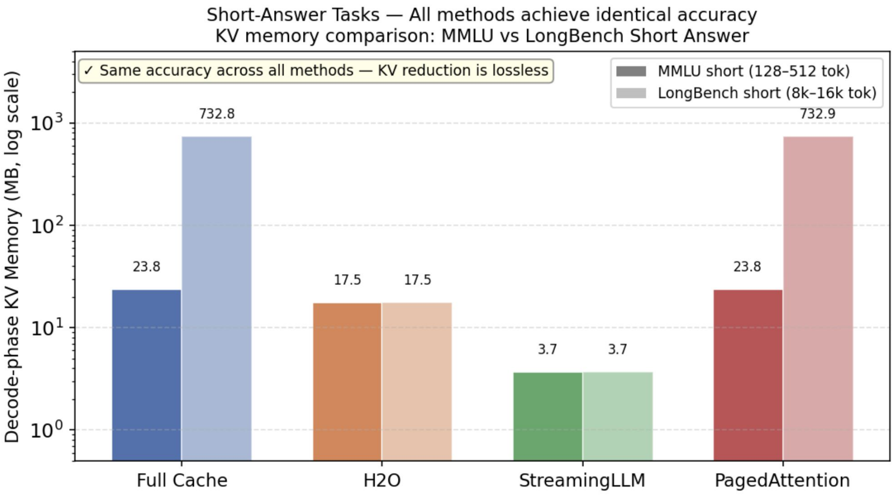
- latency
    - ttft_ms
    - decode_latency_ms: total_time_ms - ttft_ms
    - total_time_ms
- Throughput
    - throughput_tok_s: generated tokens / total time
- Memory
    - peak_kv_memory_mb
    - prefill_kv_memory_mb
    - n_oom
- quality
    - MMLU/LongBench -> accuracy
    - GovReport -> ROUGE-L

**Concurrent Serving measures additional**

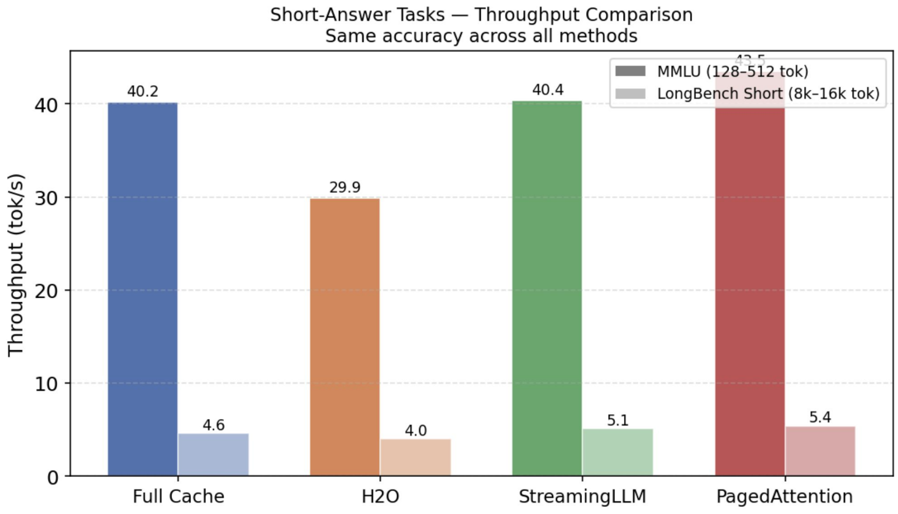
- request throughput req/s
- p50/p95/p99 TTFT & E2E latency
- service time
- peak/avg/p95 KV block utilization
- per-bucket quality

# Results — Short-Answer Tasks

- *Quality unchanged → controlled setting confirmed*
- *Memory reduction confirmed (StreamingLLM 198×, H2O 42× on LongBench)*
- *System cost differences emerge (H2O 25% lower throughput)*



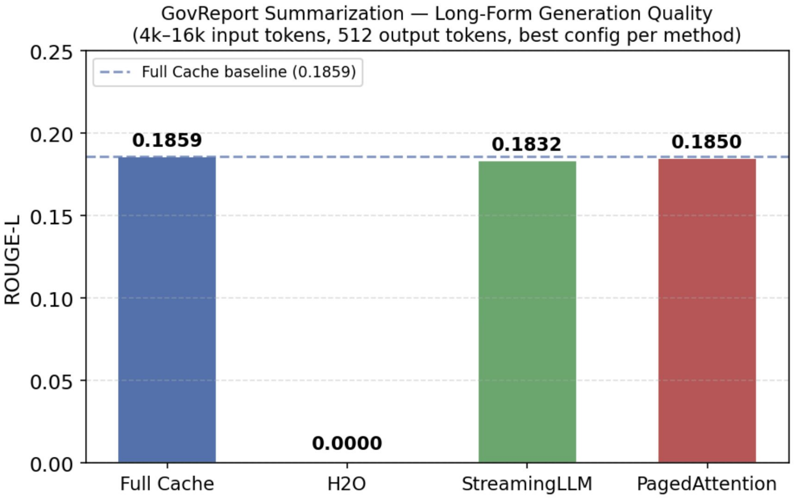

StreamingLLM: 6.4× less KV than Full Cache on MMLU | 198× less on LongBench | Zero accuracy loss




H2O is ~25% slower due to unfused eager attention kernel required for eviction scoring

# Results — Long-Form Generation

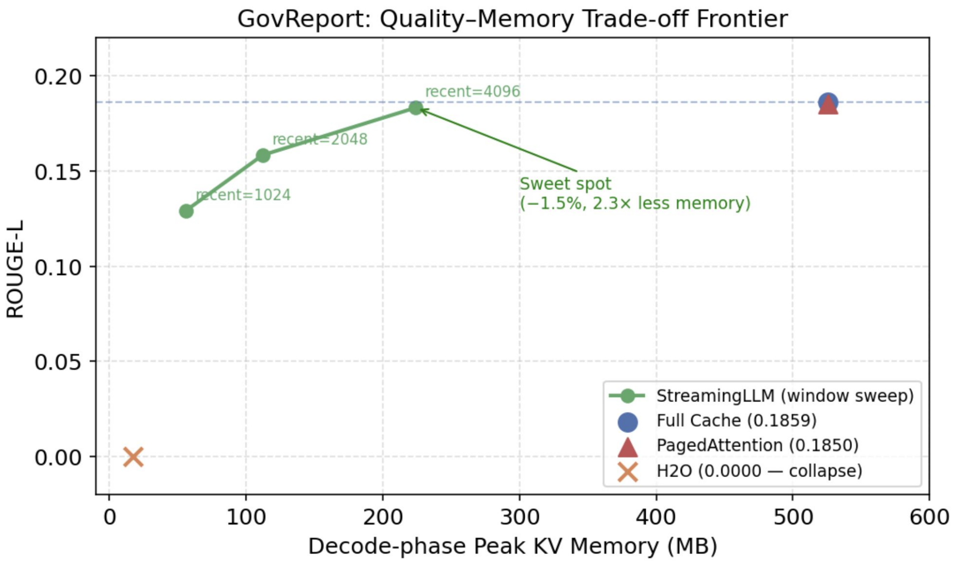
- *GovReport: 512 output tokens, 4k–16k input*
- *H2O: ROUGE-L = 0.000 at all budgets (128 to 2048 tokens)*
- *StreamingLLM (recent=4096): ROUGE-L = 0.1832 vs baseline 0.1859 (−1.5%)*
- *PagedAttention matches Full Cache*




H2O: ROUGE-L = 0 at all budgets tested (128–2048 tokens)
StreamingLLM (recent=4096): −1.5% vs baseline with 2.3× less KV memory

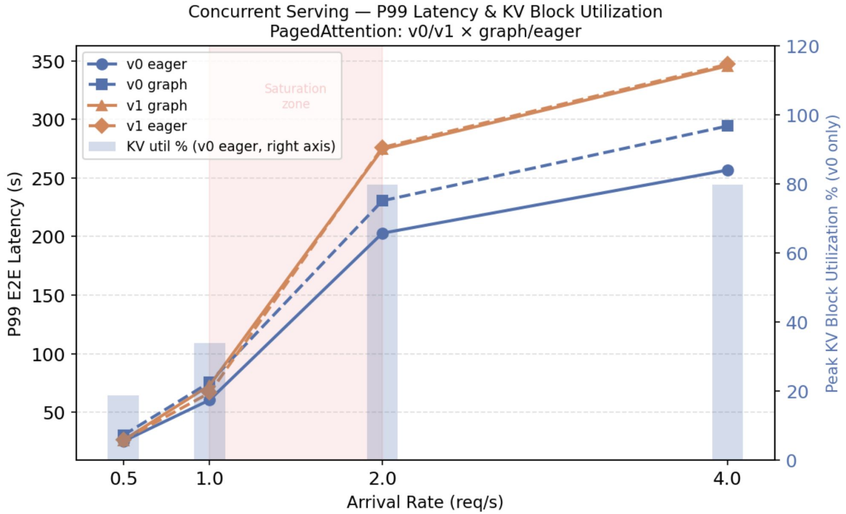
# Results — Quality-Memory Trade-off

- *Window size is a tunable knob*
- *recent=1024 ⟶ −30% ROUGE-L, 9.3× less memory*
- *recent=2048 ⟶ −15% ROUGE-L, 4.7× less memory*
- *recent=4096 ⟶ −1.5% ROUGE-L, 2.3× less memory ⟵ sweet spot*



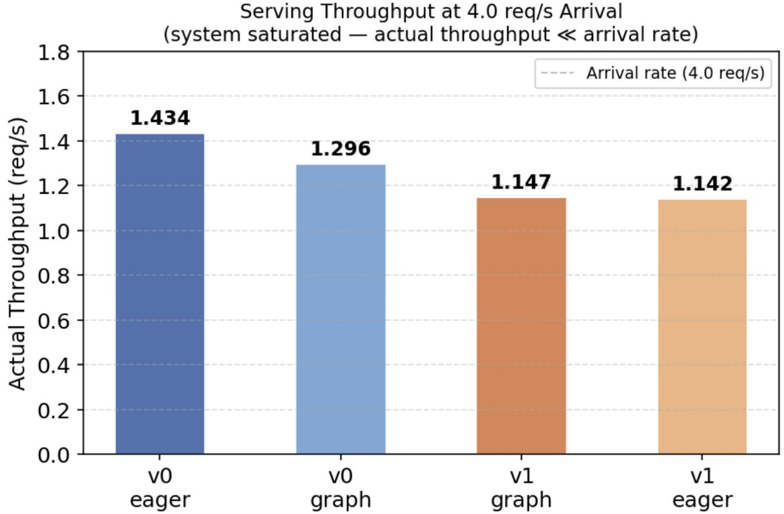
# Results — Serving Results: Latency & KV Pressure

- *System saturates between 1.0 and 2.0 req/s*
- *v0_eager: best throughput (1.43 req/s), lowest P99 E2E*
- *KV block utilization peaks at 79.7% at saturation (v0 only)*
- *Low load (0.5 req/s): 18.7% KV util — headroom available*
- *High load (2.0+ req/s): 79.7% KV util — memory becomes bottleneck*
- *Quality identical across all configs*



# Results — Quality-Memory Trade-off

- *Graph vs Eager: variable-length batches miss CUDA graph shapes → overhead without benefit*
- *v0 vs v1: chunked prefill in v1 (2048 tok/chunk) — 8 steps per long request vs 1*
- *Limitation: streaming_llm_vllm concurrent serving failed — CUDA race condition in block manager patches*
- *Takeaway: for mixed-length single-GPU serving, v0 eager + PagedAttention is the right config*




v0 eager is 24% faster than v1 — CUDA graphs hurt variable-length batches; v1 chunked prefill adds scheduling overhead

# Lessons Learned

*Time budget: 1 minute. What surprised you? What didn't work? Be honest.*

## What worked

- **StreamingLLM's sliding window scales gracefully.**

  Increasing the recent window from 1024 to 4096 tokens produced a smooth quality-memory Pareto curve, landing within 1.5% of full cache ROUGE-L at 2.3x less memory. Simple, predictable, easy to tune.

- **PagedAttention (v0 eager) is the best serving configuration.**

  Under a heterogeneous workload with extreme length variance (128 to 32k tokens), v0 eager hit 1.416 req/s at saturation, 10% higher than v0 with CUDA graphs. Disabling graphs avoided the shape-mismatch overhead that hurts mixed-length traffic.

- **Shared evaluation pipeline kept results comparable.**

  Every method returned the same `MethodOutput` schema, used the same prompts, and ran against the same trace (seed=42). This let us isolate KV cache behavior without worrying about differences in data loading or metric computation.

# Next Steps

*Time budget: 30 seconds. Be specific — vague "more experiments" is not next steps.*

1. **Larger GPU to properly evaluate H2O**

   H2O needs raw attention scores to pick heavy hitters, which forces eager (unfused) attention since FlashAttention doesn't expose scores. Eager attention is $O(n^2)$ memory during prefill, so on a single 48GB A6000 we couldn't push budgets high enough on 4k-16k inputs to give H2O a fair shot, hence ROUGE-L = 0 on GovReport.

   An 80GB H100 or H200 would fit the eager prefill at meaningful budgets and actually test whether H2O's eviction policy holds up on long generation.

2. **LongBench explain accuracy was low across all methods.**

   Even the full cache baseline only hit 27.3% on explain-format prompts (vs 40.9% on short-answer). Makes it harder to draw clean conclusions about eviction quality on that benchmark.

3. **Multi-GPU tensor parallelism serving benchmark**

   All results are single A6000 (48GB), but v1 was built for multi-GPU. Running 2x/4x with TP would show if v1 pulls ahead with its intended parallelism, how KV pressure shifts across devices, and let us push past 34k context.

# Teamwork & Contributions

*Time budget: 30 seconds. Who did what. Required slide — graded individually.*

| Member | Primary contributions | Tools / artifacts |
|---|---|---|
| Ting-Feng (Fred) | Streaming LLM, Coordinating/Setting Up | StreamingLLM HF + vLLM implementations, window size sweep experiments |
| Hung-Kai (Will) | Full KV cache (HF), Coordinating/Setting Up | Full cache baseline, shared BaseMethod interface, data pipeline, evaluation metrics |
| Sripad | H20 | H2O eviction method, attention score debugging and KV cache profiling |
| Kane | PagedAttention (vLLM) | PagedAttention vLLM integration, concurrent serving benchmark, Slurm infrastructure |

*Collaboration: regular [weekly / twice-weekly] meetings via [Zoom / in person], shared [Slack / Discord] channel, code review on every PR.*

# Thank you

*Questions?*

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Tracking: